TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

GPT-6 Astra takes on PCB layout

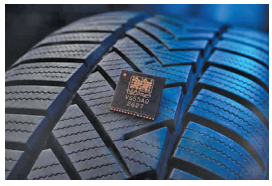
OpenAI has introduced GPT-6 Astra, a model for software engineering, scientific research, cybersecurity, and professional tasks. It uses pre-training, reinforcement learning, and alignment to improve task performance. According to OpenAI, GPT-6 Astra scored 57.9% on Terminal-Bench 4.0, which evaluates terminal-based tasks involving software engineering, system configuration, and data analysis. This compares with 55.8% for Claude Fable 5.1 and 37.3% for GPT-5.6 Sol. The model can also use KiCad to convert an electronic schematic into a PCB layout by placing components and routing copper connections. Automating parts of this process could reduce the time engineers spend on layout and allow them to focus on developing, optimising, and testing designs.



OpenAI
https://openai.com

Tyre-embedded chip cuts braking distance by 12%

POLYN Technology has rolled out VibroSense TMS, a solution that detects tyre-road friction to help vehicle systems improve braking and safety. Installed inside a tyre, the VibroSense chip processes accelerometer data to estimate the peak friction coefficient (PFC) between the tyre and the road surface under each wheel. Tyre manufacturers, vehicle original equipment manufacturers (OEMs), chassis suppliers, advanced driver assistance system (ADAS) developers, and autonomous driving technology providers can now test the sensor nodes. According to the company, a third-party test showed a 12% reduction in stopping distance. When braking from 40 km/h on wet basalt pavement, the vehicle stopped in 55.5 metres, compared with 63.2 metres without the system, a saving of 7.7 metres.



POLYN Technology
https://polyn.ai

An acoustic camera that hears leaks and electrical faults

SONOTEC has expanded its preventive maintenance portfolio with the SONASCREEN Sensor, a stationary acoustic camera that monitors industrial equipment for leaks and electrical partial discharges. The sensor uses 176 digital MEMS microphones, an optical camera, and a thermal imaging camera to detect and locate acoustic anomalies. Unlike the mobile acoustic cameras used during inspections, the sensor is designed to monitor plant areas continuously. It can be installed on equipment or integrated into mobile systems such as robots. Edge computing allows it to perform measurements, store data locally, and provide results for processing. Its applications include early detection of compressed air leaks and partial discharges in electrical systems. This helps maintenance teams assess air losses and identify electrical issues.



SONOTEC
https://www.sonotecusa.com

CHERI MCU brings hardware-enforced memory safety

SCI Semiconductor has announced silicon availability of its ICENI microcontroller (MCU) family, which uses the CHERI architecture to provide hardware-enforced memory safety. The company describes it as the first commercial implementation of CHERI in silicon. Memory-safety vulnerabilities account for about 70 per cent of reported security flaws in embedded software. ICENI addresses them through CHERI's capability model, which replaces traditional pointers with bounded capabilities that define accessible memory regions and permissions, so software components cannot reach beyond their assigned limits. The family also isolates components from one another, preventing faults and security breaches from spreading across a system. Unlike traditional memory protection units (MPUs), this approach supports component-level isolation without relying on broad memory regions.



SCI Semiconductor
https://www.scisemi.com

AI wafer reader improves chip tracking

Cognex Corporation has launched the In-Sight 1750 Series, a wafer and panel identification system designed to help semiconductor manufacturers track wafers, reduce production interruptions, and maintain throughput. The system combines artificial intelligence (AI), imaging, and illumination to read wafer and panel markings,

